

BATTA ADITYA YADAV

Dallas, TX | 857-334-5061 | aditya.yadavbatta123@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

5+ years delivering AI, ML, and NLP solutions, driving measurable business impact, with certifications from Microsoft, Google, and Databricks

WORK EXPERIENCE

Northeastern University Data Lab, Boston, USA

Feb 2024 –May 2025

Research Assistant

- Designed a multilingual AI chatbot using **LangChain** and **RAG**, achieving 78% response accuracy on NEU datasets
- Architected a **multi-agent asynchronous** workflow enabling parallel task execution and reasoning across AI agents
- Integrated **PyTorch** and TensorFlow for **multimodal fine-tuning** with Hugging Face transformers on datasets
- Transformed unstructured data into optimized SQL schemas, enabling 99% query accuracy with **MLflow** tracking
- Developed end-to-end **MLOps** pipelines using Airflow, Docker, and GitHub Actions for CI/CD automation
- Trained **AI agents** using **Reinforcement Learning** on **Hugging Face** datasets for query optimization & **Observability**
- Deployed scalable production workloads on GCP using **Kubernetes**, ensuring reliable model serving
- Evaluated multiple LLMs by analyzing **latency**, **processing time**, and response **quality** to optimize production

Johnson & Johnson, MA, USA

Jun 2024 –Dec 2024

Delivery Operations Supply Chain-Data Scientist

- Designed interactive **Power BI** dashboards tracking daily space performance of returns for stakeholders and team
- Designed and deployed a secure AI-powered **SQL** chatbot connecting large scale databases to **Vertex AI**, **GCP** and **J&J GenAI** for order tracking, reporting, and direct **Excel/Power BI** integration with HPC performance
- Optimized MedTech financial transactions with GenAI **VBA** and Excel **macros**, faster than VLOOKUP thereby reducing data prep time by 35% and enabling revenue and cost risk analysis
- Applied **predictive modeling** in Python to optimize transportation routes and schedules, yielding **\$6M** in CIP savings
- Developed **ARIMA** forecasting models on inventory and space-utilization data, increasing predictive accuracy by 32%
- Mentored 2-4 junior staff during on **Databricks** integration with GenAI and Power BI for financial data analysis

Vitel Global, Hyderabad, India

Jun 2019– Jun 2023

ML Engineer

- Developed scalable web applications using **C++**, **JavaScript**, and **React.js**, and built **Python ETL** and **SQL** pipelines on **AWS EC2** to process millions of telecom records, improving efficiency by **47%**
- Executed advanced data mining and analytical modeling on **50M+ VoIP payment transactions**, identifying fraud and revenue leakage and recovering **\$2.3M+** in annualized revenue
- Implemented **ML classifiers** (Random Forest, **XGBoost**) for fraud detection on large web-call datasets
- Designed deep learning structures using **CUDA**, **TensorFlow**, and **JAX** for employee facial-ID detectors
- Streamlined real-time data pipelines using Apache **Kafka**, increasing system reliability for high-volume streaming data

SELECTED PROJECTS

Open-Source Python Library: Unstructured data to Relational Database – QuerySUTRA 0.5.0

Nov 2025

- Developed a python library using **Lang Chain**, **AI agents**, **embeddings**, **semantic search**, **snorkel** and **FAISS Vector**

Financial Stock Analytics: Multilingual GenAI Chatbot & Video Intelligence - MCP Server

Aug 2025

- Build a GenAI chatbot using **NLP**, **RAG pipelines**, **FAISS**, **FastAPI**, **MCP**, and **LLMs** for financial summarization

Intelligent Agentic AI Web Scraper & Real-Time Data Intelligence – FireCrawl API

Mar 2025

- Designed an autonomous AI web scraper using **FireCrawl API**, and vector databases for structured real-time insights

Multi-Agentic AI Application: Rental Car Services Optimization (USA) - CrewAI

Feb 2025

- Built a CrewAI model using **Ollama**, **GCP**, and real-time **Kayak** data for demand forecasting and pricing optimization

TECHNICAL SKILLS

Programming : Python, MATLAB, Java, Spark, **Pandas**, **Databricks** Agentic AI

Databases : MySQL, MongoDB, Big Query, AWS(S3), Docker, **GCP**, Azure, Chroma DB, Faiss DB

ML : Regression, **Predictive Modelling**, Clustering, Decision Modelling, PC, **Kubernetes**

Gen-AI : LLM's, RAG, Agentic AI, NLP, Deep Neural Networks, Word Embeddings, CNN's, RNN's, VAE

NLP : Transformers, BERT, GPT-style models, Prompt Engineering, Chat Summarization, NLTK

EDUCATION

Northeastern University, Boston, MA, USA

Aug 2025

Candidate for Master of Science in Data Analytics Engineering

GPA: 3.8

Related courses: Machine Learning Operations, Gen AI, Neural Networks and Deep, Data Mining, Data Management for Analytics, Computational and Visualization, Foundations of Data Analytics, **GenAI**, **Economic Decision Making**